

Wearable Golf Device

Part Number: WGD-6826

REV: A

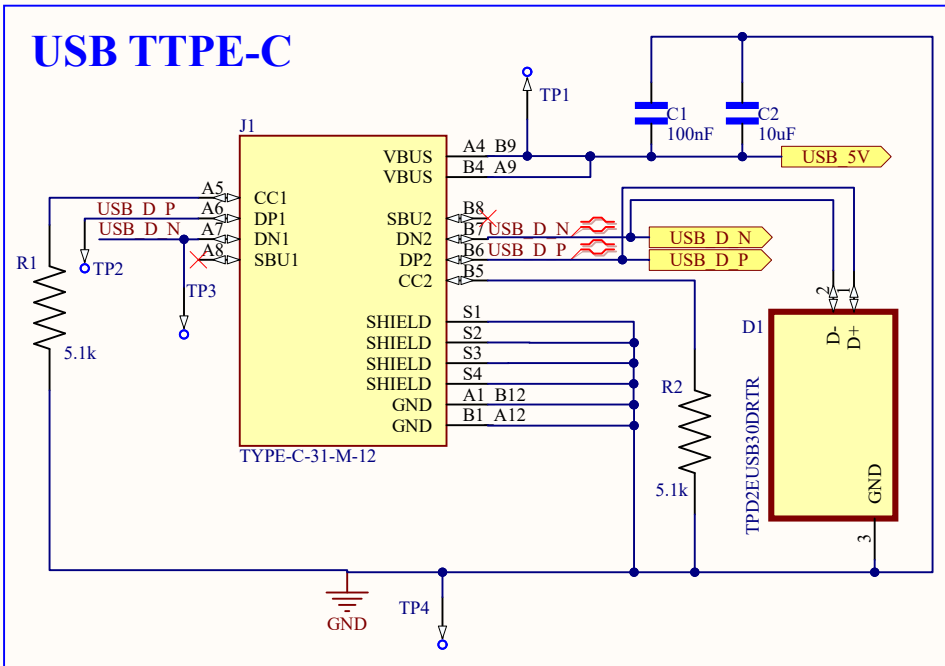
Schematics Draft: 1

Contents

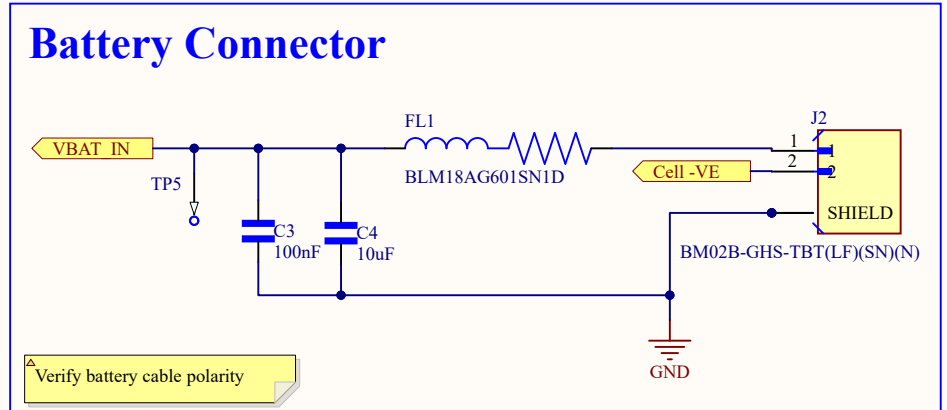
Connectors.....	2
Power Supply Unit.....	3
Display Unit.....	4
Selection Keys.....	5
Micro Control Unit.....	6
GPIO Mapping.....	7
Bills of Material.....	8

CONNECTORS

USB TTPE-C



Battery Connector

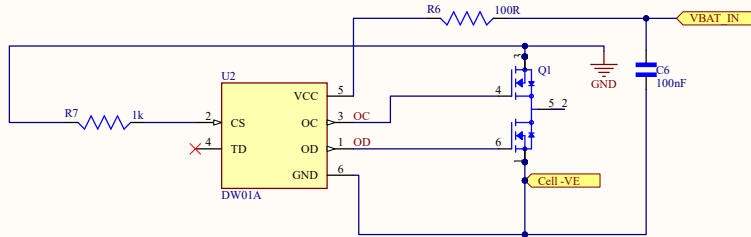


Verify battery cable polarity

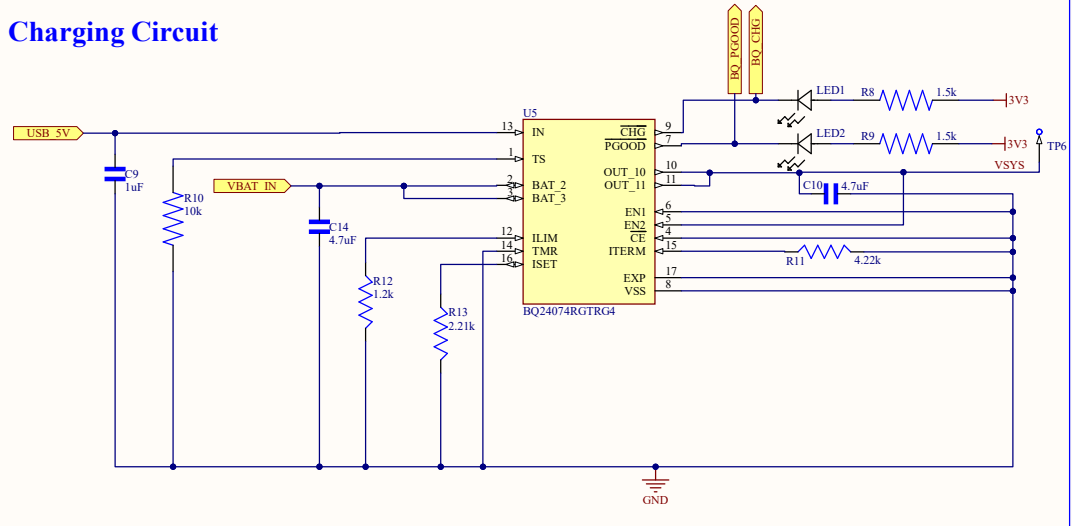
Title			Wearable Golf Device	
Size	Number	Revision		
A	WGD-6826	A		
Date:	6/21/2026	Sheet of	2 of 8	
File:	C:\Users\...\Connectors.SchDoc	Drawn By:	Engr. M TAIMOOR	

POWER SUPPLY UNIT

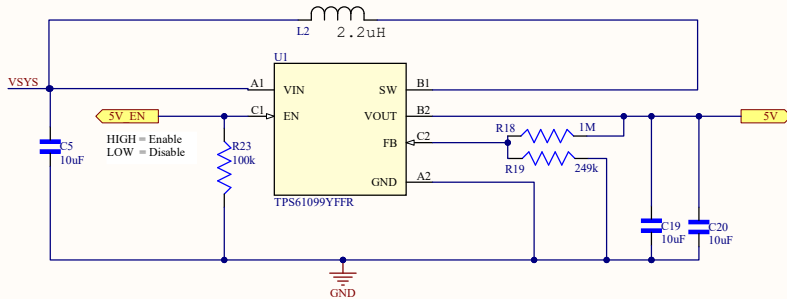
Battery Protection Circuit



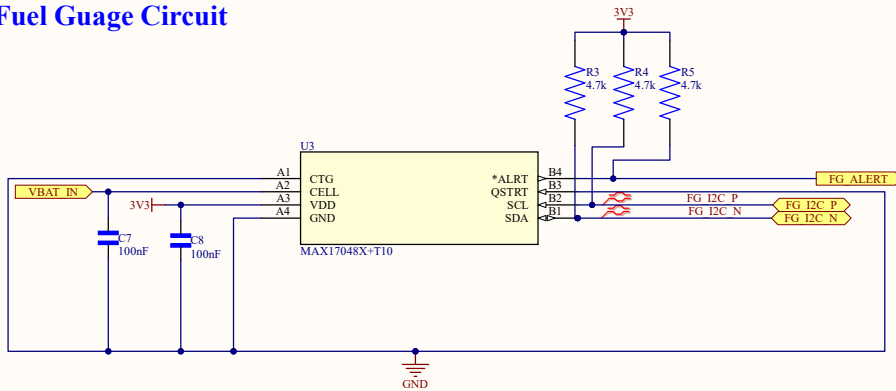
Charging Circuit



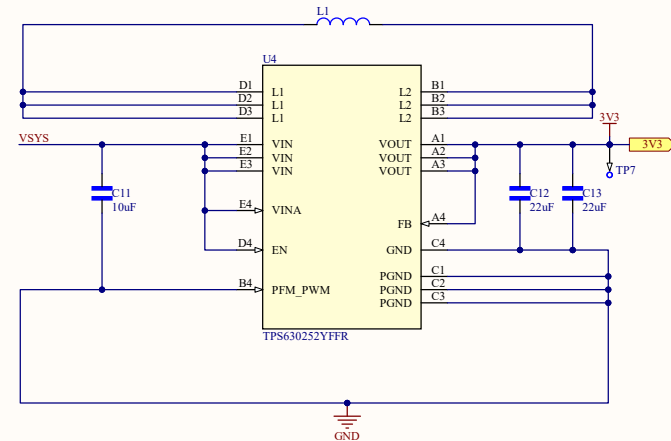
5V Regulator Circuit



Fuel Gauge Circuit

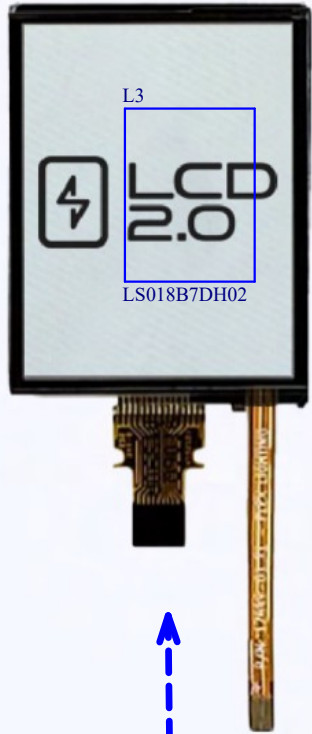


3V3 Regulator Circuit

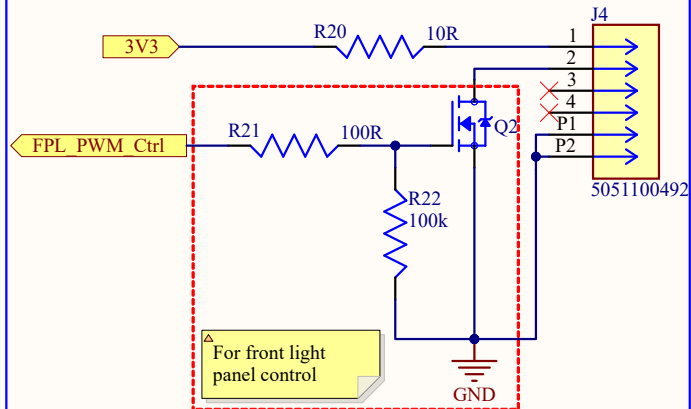


Title		Wearable Golf Device	
Size	Number	Revision	
A3	WGD-6826	A	
Date:	6/21/2026	Sheet of	3 of 8
File:	C:\Users\...Power Supply Unit.SchDoc	Drawn By:	Engr. M.TAIMOOR

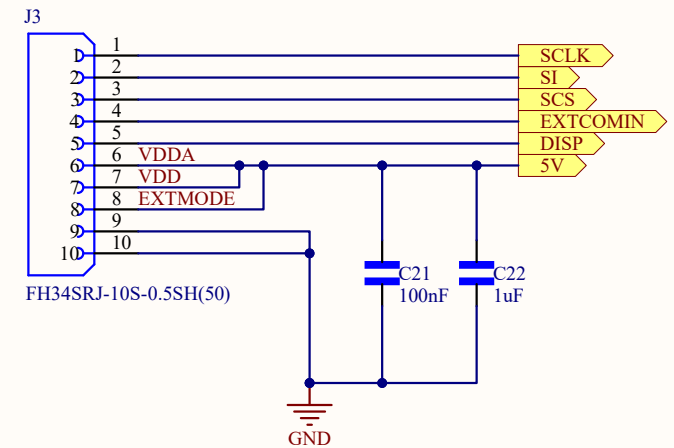
DISPLAY UNIT



FLP Connector



LCD Connector

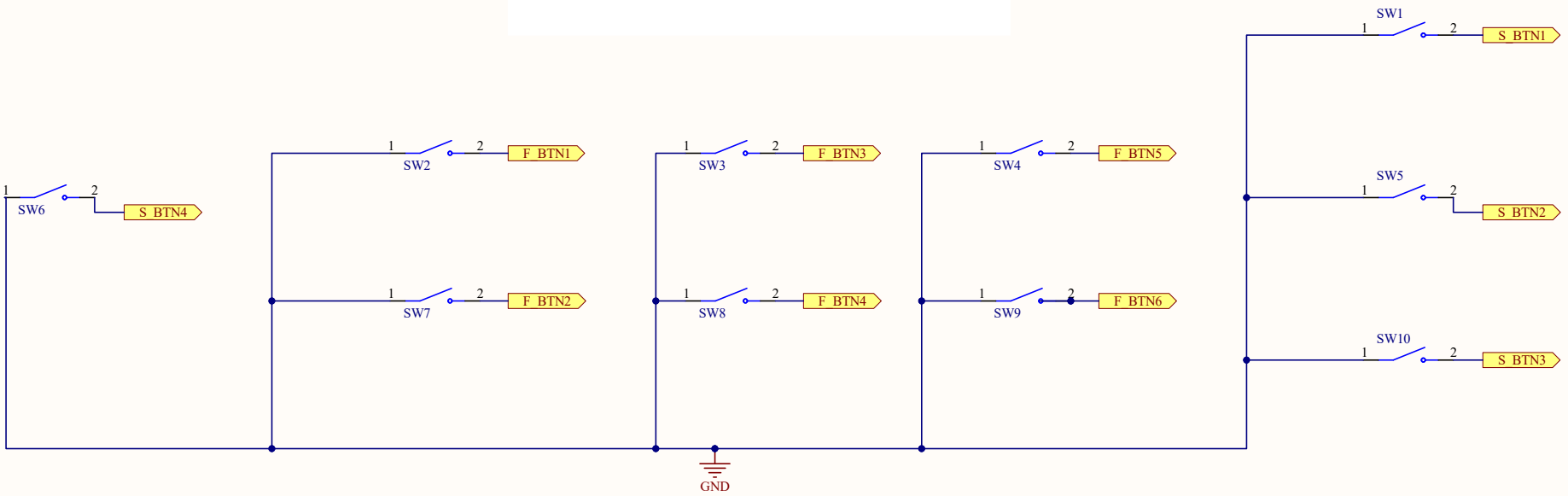


Title			
Wearable Golf Device			
Size	Number	Revision	
A	WGD-6826	A	
Date:	6/21/2026	Sheet of	4 of 8
File:	C:\Users\...\Display Unit.SchDoc	Drawn By:	Engr. M TAIMOOR

SELECTION KEYS

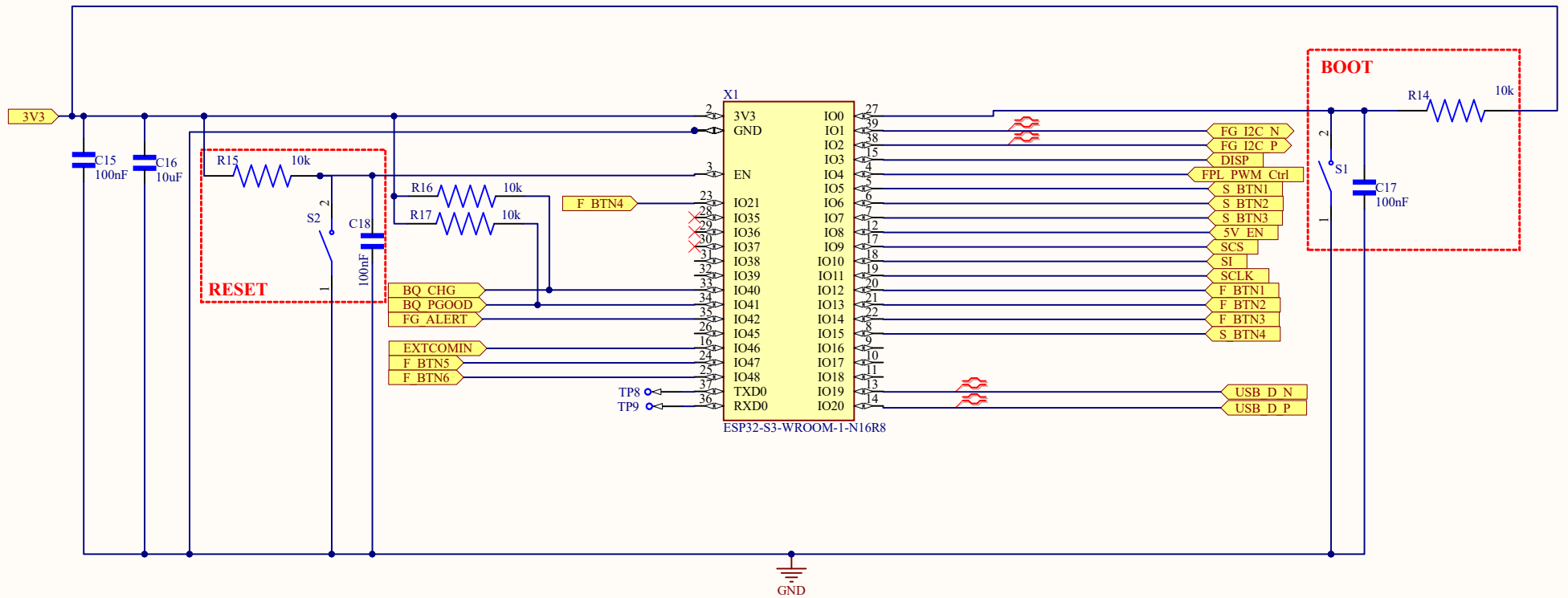
FRONT BUTTON IMPLEMENTATION NOTE

- Front keypad buttons use metal dome contacts (HYP-600-315S-160)
- Center pad (Pin 1) is connected to the ESP32 GPIO input.
- Outer ring (Pin 2) is connected to GND.
- Buttons are configured as ACTIVE-LOW inputs.
- ESP32 internal pull-up resistors shall be enabled in firmware.
- Button released: GPIO reads HIGH.
- Button pressed: metal dome shorts center pad to outer ring, GPIO reads LOW.
- Recommended firmware configuration: INPUT_PULLUP.



Title			Wearable Golf Device	
Size	Number			Revision
A	WGD-6826			A
Date:	6/21/2026	Sheet of	5 of 8	
File:	C:\Users\...\Selection Keys.SchDoc	Drawn By:	Engr. M TAIMOOR	

MICRO CONTROL UNIT



Title			
Wearable Golf Device			
Size	Number	Revision	
A4	WGD-6826	A	
Date:	6/21/2026	Sheet of	6 of 8
File:	C:\Users\...\MCU.SchDoc	Drawn By:	Engr. M TAIMOOR

GPIO MAPPING

MCU: ESP32-S3-WROOM-1-N16R8

BUTTON MAPPING

Ser	Name	Location	Function	GPIO
1.	F_BTN1	Front Button 1	User Defined	GPIO 12
2.	F_BTN2	Front Button 2	User Defined	GPIO 13
3.	F_BTN3	Front Button 3	User Defined	GPIO 14
4.	F_BTN4	Front Button 4	User Defined	GPIO 21
5.	F_BTN5	Front Button 5	User Defined	GPIO 47
6.	F_BTN6	Front Button 6	User Defined	GPIO 48
7.	S_BTN1	Left Button 1	User Defined	GPIO 5
8.	S_BTN2	Left Button 2	User Defined	GPIO 6
9.	S_BTN3	Left Button 3	User Defined	GPIO 7
10.	S_BTN4	Right Button 1	User Defined	GPIO 15

Note: All buttons are active-low and require ESP32 internal pull-up resistors enabled in firmware.

DISPLAY: LS018B7DH02

DISPLAY MAPPING

Ser	Name	Signal	Function	GPIO
1.	SCLK	LCD_SCLK	SPI Clock	GPIO 11
2.	SI	LCD_MOSI	SPI Data	GPIO 10
3.	SCS	LCD_CS	Chip Select	GPIO 9
4.	DISP	LCD_DISP	Display Enable	GPIO 3
5.	EXTCOMIN	LCD_EXTCOMIN	COM Inversion	GPIO 46

FLP MAPPING

6.	FPL_PWM_Ctrl	FLP_PWM	Front Light Brightness Control	GPIO 4
----	--------------	---------	--------------------------------	--------

Note: Front light driven through 2N7002 MOSFET.

POWER MANAGEMENT MAPPING

Fuel Gauge: MAX17048X+T10

Ser	Name	Signal	Function	GPIO
1.	FG_ALERT	FG_ALERT	Fuel Gauge Alert	GPIO 42
2.	FG_I2C_P	I2C_SCL	Serial Clock	GPIO 1
3.	FG_I2C_N	I2C_SDA	Serial Data	GPIO 2

Charging IC: BQ24074RGTRG4

4.	BQ_CHG	-	Charger Status	GPIO 40
5.	BQ_PGOOD	-	Power Good Status	GPIO 41

USB: TYPE-C-31-M-12

6.	USB_D_N	USB Data	-	GPIO 19
7.	USB_D_P	USB Data	-	GPIO 20

BOOT/RESET

8.	ESP32_PIN27	BOOT	-	GPIO 0
9.	ESP32_PIN3	RESET	-	EN

5V Regulator: TPS61099YFFR

10.	5V_EN	EN	Control 5V for LCD HIGH = Enable LOW = Disable	GPIO 8
-----	-------	----	--	--------

BILLS of MATERIAL

Designator	Quantity	MPN	LCSC Part #	Description
C1, C3, C6, C7, C8, C15, C17, C18, C21	9	CL03A104KQ3NNNC	C160831	CAP CER 100nF 6.3V X5R 0201
C2, C5, C16, C19, C20	5	CL10A106KP8NNNC	C19702	CAP CER 10uF 10V X5R 0603
C4, C11	2	CL10A106MQ8NNNC	C1691	10 μ F $\bar{A}$ ±20% 6.3V Ceramic Capacitor X5R 0603 (1608 Metric)
C9, C22	2	CL05A105KP5NNNC	C14445	1 $\bar{A}$ μ F $\bar{A}$ ±10% 10V Ceramic Capacitor X5R 0402 (1005 Metric)
C10, C14	2	CL05A475MQ5NRNC	C307423	CAP CER 4.7uF 6.3V X5R 0402
C12, C13	2	CL10A226MQ8NRNC	C59461	CAP CER 22uF 6.3V X5R 0603
D1	1	TPD2EUSB30DRTR	C97502	2-channel ESD solution for SuperSpeed USB 3.0 interface 3-SOT-9X3 -40 to 85
FL1	1	BLM18AG601SN1D	C19330	600 Ohms @ 100 MHz 1 - Ferrite Bead 0603 (1608 Metric) 500mA 380mOhm
J1	1	TYPE-C-31-M-12	C165948	USB Connectors 24 Receptacle 1 8.94*7.3mm RoHS
J2	1	BM02B-GHS-TBT(LF)(SN)(N)	C161690	Connector Header Surface Mount 2 position 0.049 (1.25mm)
J3	1	FH34SRJ-10S-0.5SH(50)	C324723	10 Position FFC, FPC Connector Contacts, Top and Bottom 0.020 (0.50mm) Surface Mount, Right Angle
J4	1	2328702-4	C324723	4 Position FPC Connector Contacts, Top and Bottom 0.020 (0.50mm) Surface Mount, Right Angle
L1	1	YSMC160865H-1R0MG	C46961960	FIXED IND 1uH 0603
L2	1	CC322522A-1R0K	C21160813	FIXED IND 2.2uH
L3	1	LS018B7DH02		Azumo part number 12616-03 is a 1.8" monochrome, transfective, active-matrix liquid crystal display (LCD) with integrated front light panel (FLP) in an "-03" configuration
LED1	1	XL-0201SURC	C3646923	LED RED 0201 SMD
LED2	1	XL-0201UBC	C3646921	LED BLUE 0201 SMD
Q1	1	FS8205A	C2830320	MOSFET 20V 9A SOT-23-6
Q2	1	2N7002	C124475	Mosfet Array 2 N-Channel (Dual) 60V 300mA 500mW Surface Mount PG-SOT363-6
R1, R2	2	RC0201DR-075K1L	C851767	RES 5.1k Ω ±0.5% 50mW 0201
R3, R4, R5	3	RC-01W4701FT	C320923	RES 4.7k Ω ±1% 50mW 0201
R6, R21	2	RC0402FR-07100RL	C106232	100 Ohms $\bar{A}$ ±1% 0.063W, 1/16W Chip Resistor 0402 (1005 Metric) Moisture Resistant Thick Film
R7	1	RC0402FR-071KL	C106235	RES 1k Ω ±1% 62.5mW 0402
R8, R9	2	RC0201FR-071K5L	C155737	RES 1.5k Ω ±1% 50mW 0201
R10, R14, R15, R16, R17	5	ERJ-1GNF1002C	C717002	10 kOhms 1% 0.05W, 1/20W Chip Resistor 0201 (0603 Metric) Automotive AEC-Q200 Thick Film
R11	1	ERJ2RKF4221X	C416537	RES 4.22k Ω ±1% 100mW 0402
R12	1	RC0402FR-071K2L	C138040	RES 1.2k Ω ±1% 62.5mW 0402
R13	1	RC0402FR-072K21L	C138019	RES 2.21k Ω ±1% 62.5mW 0402
R18	1	AC0402FR-071ML	C144787	RES 1M Ω ±1% 62.5mW 0402
R19	1	ERJ2RKF2493X	C2076822	RES 249k Ω ±1% 100mW 0402
R20	1	RC0402FR-0710RL	C138066	RES 10 Ω ±1% 62.5mW 0402
R22	1	RC0402FR-07100KL	C60491	RES 100k Ω ±1% 62.5mW 0402
S1, S2	2	TS-1088-AR02016	C720477	50mA 4mm 100M $\bar{I}$ @ 100,000 Times 12V 160gf 3mm 2mm Round Button Vertical welding SPST SMD Tactile Switches ROHS
SW1, SW5, SW6, SW10	4	GT-TC035B-H0195-L3	C48888560	SWITCH TACTILE SPST 250gf SMD R/A
SW2, SW3, SW4, SW7, SW8, SW9	6	600-315S-160	C256197	METAL DOME FOR TACT SWITCH
U1	1	TPS61099YFFR	C91091	0.7Vin Synchronous Boost Converter with 800 nA Ultra-Low Quiescent Current 6-DSBGA -40 to 85
U2	1	DW01A	C2927799	One Cell Lithium-ion/Polymer Battery Protection IC
U3	1	MAX17048X+T10	C7497002	3 $\bar{A}$ μ A 1-Cell/2-Cell Fuel Gauge with ModelGauge
U4	1	TPS630252YFFR	C2070606	4A switch Single-Inductor Buck-Boost Converter in WCSP 20-DSBGA -40 to 125
U5	1	BQ24074RGTRG4	C54313	Standalone 1-cell 1.5-A linear charger, Power Path, 4.2-V VBAT, 10.5-V OVP and 4.4-V VOUT 16-VQFN -40 to 85
X1	1	ESP32-S3-WROOM-1-N16R8	C2913202	Bluetooth, WiFi 802.11b/g/n, Bluetooth v5.0 Transceiver Module 2.4GHz PCB Trace Surface Mount